

Chapter 12

AI in Serving Older Adults



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1 Introduction

With the continued decline in birth rates and significant increases in life expectancy, the world is rapidly entering an era of population aging. According to the United Nations' World Population Prospects 2022 report, the global population aged 60 and above is projected to reach 2.1 billion by 2050, nearly double the 1 billion recorded in 2020 (Kalache, 2019). While challenges such as chronic diseases, physical decline, cognitive deterioration, and social isolation are inherent to the aging process, the demographic shift will significantly expand the number of older adults affected by these issues. As a result, the strain on healthcare systems, social support networks, and family caregiving structures will intensify. This growing need underscores the increasing demand for effective health management, emotional support, and cognitive maintenance services tailored to the elderly (Li, 2023).

On the other hand, aging populations are placing unprecedented pressure on elder care services due to the reduction in the younger workforce, changes in traditional family structures, and a shortage of professional caregivers (Feng, 2019; Feng & Wu, 2023). Moreover, rising costs of elder care are straining national healthcare systems, social service resources, and public policies. For instance, China's National Healthcare Security Administration recently issued new guidelines for nursing projects and policy adjustments to simplify the nursing service pricing system and improve transparency (China National Healthcare Security Administration, 2024). Additionally, Canada launched its National Housing Strategy in 2017, with one of its priorities being to improve housing outcomes for older adults (Government of

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Canada, 2024), while Japan's nursing insurance system offers various levels of care services for the elderly (Yamada & Arai, 2020). Singapore, meanwhile, has introduced measures to raise the retirement age and provide skill development opportunities to help older individuals re-enter the workforce (Sun & Usui, 2023). In the U.S., a comprehensive national framework is also underway. The Administration for Community Living, through the Interagency Coordinating Committee (ICC) on Healthy Aging and Age-Friendly Communities, released a Strategic Framework for a National Plan on Aging in 2024 (Administration for Community Living, 2024). This vision promotes coordinated public-private partnership and cross-sector collaboration—including federal agencies, industry, philanthropy, academia, and older adults—and calls for initiatives in age-friendly communities, aligned health and supportive services, coordinated housing, and increased access to long-term care. It emphasizes person-centeredness and respect for older adults' dignity and autonomy, and aims to dismantle ageism and ableism—offering a vital example of how policy frameworks can integrate both humanistic and technological considerations in preparing for population aging in a developed country context.

In recent years, the rapid advancement of artificial intelligence (AI) technology, especially in the healthcare sector, has significantly improved the accuracy of disease diagnosis and the personalization of treatment plans. For example, AI-assisted diagnostic systems analyze medical images and patient data to quickly identify abnormalities, aiding doctors in formulating more effective treatment plans, thereby providing more precise medical services. Furthermore, AI-driven intelligent dialogue systems offer personalized, user-friendly consultation services for patients (Quazi et al., 2024). In terms of cognitive interventions, AI-powered cognitive training platforms can provide customized brain exercises for the elderly, helping slow the progression of cognitive decline (Faria et al., 2024). Compared to traditional methods, AI offers significant efficiency gains. It can tailor care plans to each elderly individual's unique health status and personalized needs, while also providing continuous, dynamic support throughout the care process.

Although existing research has explored the application of AI in elderly care, comprehensive discussions on systematic evaluation, cultural adaptability, and ethical design remain scarce (Asan & Choudhury, 2021; Lukkien et al., 2023). Therefore, this chapter will delve into the practical effects and potential of AI in elderly care, focusing on its innovative applications in health monitoring, social engagement, cognitive support, and aging in place. Additionally, the chapter will include an exploration of challenges related to technical adaptability, ethical concerns, and cultural differences—not only as barriers to AI implementation, but as key areas where social work can play a transformative role. By addressing these challenges, social workers can help ensure that AI technologies support human dignity, equity, and culturally responsive care for older adults.

2 Application of AI in Elderly Care

2.1 AI in Health Monitoring for Older Adults

In the field of elderly care, AI technology has become a central application in health monitoring. As the elderly population grows and health needs become more diversified, AI technologies, such as smart wearable devices and mobile applications, enable real-time monitoring and data collection of vital signs in older adults (Awotunde et al., 2022; Sabry et al., 2022). These data are processed and analyzed through cloud-based platforms, using AI algorithms for disease prediction and risk assessment, thereby providing real-time, convenient, personalized, and efficient health monitoring services. AI plays a crucial role in early disease detection, chronic disease management, accident prevention, and personalized health interventions.

2.1.1 Early Disease Detection

AI has significant potential for early disease detection, particularly in predicting and monitoring atrial fibrillation, where its applications are transforming medical diagnostics and management (Salvi et al., 2024; Yao et al., 2023). By analyzing physiological data from wearable devices, such as heart rate, blood pressure, and oxygen saturation, AI can detect potential disease risks. For example, using wavelets and AI on EEG data, atrial fibrillation can be diagnosed early, helping to identify high-risk patients and provide timely prevention measures (Serhal et al., 2022). Deep learning techniques have demonstrated excellent performance in the early detection of cardiovascular diseases, identifying high-risk individuals and providing decision support through data analysis (Baikuev et al., 2024).

In terms of cognitive function, wearable device activity log data can effectively support early diagnosis of dementia in high-risk older adults, improving healthcare system accessibility and efficiency (Lee & Lee, 2024). Advanced machine learning techniques, using computerized cognitive tests, motion analysis, speech processing, and brain scans, have enabled AI models to detect the risks of Alzheimer's and Parkinson's diseases with high accuracy (Grueso & Viejo-Sobera, 2021; Reddy et al., 2024). Furthermore, AI systems can closely monitor daily activities, social interactions, and emotional fluctuations in older adults to identify early signs of depression and propose appropriate interventions (Su et al., 2020).

2.1.2 Chronic Disease Management

AI plays a key role in managing chronic diseases, particularly in medication management. Older adults often need to take multiple medications for extended periods, and AI systems, through intelligent medication management systems, can provide accurate medication reminders based on treatment plans and make real-time

adjustments according to the elderly individual's medication responses (Ciampi et al., 2022; García-Vázquez et al., 2011). For example, AI can precisely adjust medication dosages based on blood glucose monitoring data, significantly reducing the risks of missed or excessive medication (Spanakis et al., 2022). Additionally, by integrating diverse health data, AI can offer personalized management plans for patients with multiple chronic diseases, such as diabetes, heart disease, and hypertension, thereby improving the overall health of older adults (Wu et al., 2021; Majnarić et al., 2021).

2.1.3 Prevention of Accidental Events

AI technology plays a critical role in mitigating accidental events among older adults, such as fall prevention. For example, fall detection systems that integrate AI analysis can monitor the gait and posture of older individuals in real time, predict fall risk, and issue alerts when necessary to reduce fall incidents and related injuries (Cedeno-Moreno et al., 2024; Kiprijanovska et al., 2020). Furthermore, by monitoring physiological data such as heart rate, blood pressure, and EEG, AI technology can provide early warnings of acute health events, such as heart attacks or strokes, and promptly notify the elderly individual, their family, or medical staff for timely intervention (Xavier et al., 2022; Islam et al., 2022). AI, combined with location-tracking devices, can monitor older adults' locations, preventing them from becoming lost or leaving a safe area, which is particularly useful for those with cognitive impairments (Moreno et al., 2015; García-Requejo et al., 2023).

2.1.4 Personalized Health Interventions

Personalized health interventions are one of AI's significant advantages. AI can develop tailored health plans for older adults based on their health data, covering diet, exercise, and medication management (Espín et al., 2016; Zhu et al., 2023). AI health assistants, leveraging real-time monitoring data and long-term accumulated data resources, can provide precise personalized health recommendations and adjust interventions accordingly to ensure efficient health management. For example, by analyzing gait characteristics, physical abilities, and subtle health trends, AI can identify early signs of health deterioration in older adults and design customized rehabilitation training plans based on these data (Ferraris et al., 2022; Kim et al., 2024). AI-IoT-based elderly healthcare programs have significantly improved the management of hypertension, diabetes, physical activity, and nutrition habits in over 97% of participants while effectively reducing frailty (Woo et al., 2024). These personalized interventions not only improve older adults' health management habits but also effectively prevent the further deterioration of health issues, enhance quality of life, and help delay the aging process.

2.2 AI in Social Engagement for Older Adults

Mental health concerns, particularly depression and anxiety, represent significant issues among older adults experiencing social isolation and loneliness. Symptoms such as persistent sadness, loss of interest in daily activities, and reduced emotional engagement severely impact their quality of life and increase demands on social work services. Social workers play a critical role in early identification, intervention, and supportive care, and AI technologies can serve as valuable tools to aid these efforts. By reducing feelings of loneliness, fostering social interaction, and providing emotional support, AI systems offer innovative strategies to address the psychosocial needs of older adults and enhance their mental well-being as populations continue to age.

2.2.1 Reducing Feelings of Loneliness

Loneliness is particularly prevalent among older adults, especially when family members are unable to visit frequently due to work, geographical distance, or other commitments, making their sense of isolation even more pronounced. Such loneliness can have serious consequences, including depression, reduced cognitive function, and poorer overall health outcomes. Social robots have emerged as a promising intervention to address social isolation and loneliness among older adults in both pandemic emergencies and aging societies (Jecker, 2020). AI technologies, particularly virtual assistants and social robots, can offer emotional support and companionship, helping alleviate loneliness among older adults (Gasteiger et al., 2021; You & Choi, 2025). Information and communication technology solutions, such as smart homes and robotic pets, can help detect and mitigate loneliness without requiring direct interaction with others (Latikka et al., 2021). These AI systems use voice recognition and natural language processing technologies to interact with older adults, offering care and companionship. For example, social robots can engage in simple conversations with older individuals, helping them cope with loneliness and providing emotional comfort (Pirhonen et al., 2020). Furthermore, AI can use emotional analysis techniques to sensitively detect fluctuations in an older adult's mood and tailor interaction strategies to their emotional needs. Through this emotional support, AI provides greater companionship and emotional connection, thereby alleviating loneliness (Qiu & Carter, 2023).

2.2.2 Promoting Social Interaction

Older adults often face difficulties in actively participating in social activities due to health problems or mobility limitations, leading to social isolation (Mehrabi & Béland, 2020). AI technology can integrate data from wearable devices and home health monitoring tools to provide comprehensive monitoring of an older adult's

health status. Moreover, AI can help create social platforms that allow older individuals to maintain connections with distant family members and friends through video calls and online gatherings, enriching their social lives. For instance, systems combining intelligent personal assistants (IPAs) and interactive televisions (iTVs) can facilitate video calls between older adults and their friends or family members, thus promoting social interactions and maintaining their social circles (Camargo et al., 2023). Additionally, AI systems can recommend suitable social activities or community groups based on an older adult's lifestyle, interests, and health status, helping them find appropriate social circles and engage in more interactions (Czaja & Ceruso, 2022; You & Choi, 2025). These carefully recommended social activities help older adults integrate into social groups, enhancing their sense of social participation, while also ensuring that these activities meet their needs and safety standards (Blusi et al., 2018).

2.2.3 Providing Emotional Support

Depressive symptoms are prevalent among socially isolated older adults and may include feelings of sadness, hopelessness, and loss of interest. These symptoms significantly affect emotional well-being and overall quality of life. In social work practice, early recognition and sustained emotional support are critical. AI technologies—particularly chatbots and social robots—are emerging as valuable tools in this regard. By continuously monitoring behavioral cues, speech patterns, and daily routines, these systems can detect early signs of depression and anxiety, enabling timely interventions by social workers and caregivers (Andrews et al., 2017; Gomes et al., 2023). For instance, AI-driven conversational agents can analyze data on sleep quality, activity levels, and social withdrawal to trigger alerts for supportive care.

Older adults often face challenges such as health issues, lifestyle changes, and social isolation, making emotional support particularly important. AI-driven social robots and chatbots are emerging as promising tools for providing emotional support and companionship to older adults, particularly those living alone (Hsu & Hsieh, 2022). These technologies include embodied conversational agents and AI robots that engage in dialogue, recommend personalized care, and assist with daily activities (Milne-Ives et al., 2020). Some systems even incorporate anthropomorphic features or mimic pets to strengthen emotional bonds (Chen & Park, 2021). In addition, AI chatbots can provide lighthearted interactions—such as singing or telling jokes—that may help manage some symptoms of dementia (Liu et al., 2018).

AI systems can also analyze real-time changes in an older adult's emotional state. For example, if sadness or apathy is detected, the AI can proactively offer comforting dialogue or suggest coping activities (Wu et al., 2023). This capability is especially important for mental health monitoring, as AI can process behavioral signals like sleep quality, physical activity, and social withdrawal to detect possible signs of depression or anxiety. When necessary, it can alert caregivers or family members for timely human support (Andrews et al., 2017; Gomes et al., 2023).

Importantly, these technologies should be viewed as assistive tools rather than replacements for human care. Social workers play an essential role in interpreting AI-generated insights, assessing emotional and contextual factors, and designing individualized support plans grounded in empathy and ethical practice. AI-assisted interventions must align with the principles of person-centered care, ensuring that technological support reinforces rather than overrides professional judgment and human relationships.

Ultimately, AI-enabled emotional care, when guided by social work values, can enhance older adults' sense of dignity, connection, and psychological security. By blending continuous, adaptive monitoring with compassionate professional oversight, these technologies have the potential to improve mental health outcomes and reduce the emotional toll of aging in isolation.

2.3 AI in Cognitive Support for Older Adults

As populations age and the proportion of older adults increases, age-related conditions such as cognitive decline—particularly cognitive frailty—have become significant factors affecting the quality of life for older adults (Wilson et al., 2013). Many older adults face cognitive challenges, including memory loss, decreased attention span, and reduced mental flexibility, which not only affect their daily lives but may also lead to mental health issues (Aajami et al., 2020). AI has shown great potential for cognitive support in older adults, particularly in three key areas: cognitive assessment, intervention, and recovery, helping them maintain cognitive function, delay cognitive decline, and ultimately improve their quality of life.

2.3.1 Cognitive Assessment

Cognitive assessment is the foundation of AI applications in cognitive support, as it helps efficiently and accurately identify early signs of cognitive impairment or decline. Traditional cognitive assessments often rely on manual evaluations or standardized tests. At the same time, AI technologies can analyze large datasets to provide more comprehensive and personalized assessments, detecting early signs of cognitive issues (Graham et al., 2019). Additionally, traditional methods often rely on single tests and lack long-term tracking and personalized analysis. In contrast, AI's advantage lies in providing real-time data and dynamic assessments, enabling continuous monitoring of the elderly's cognitive health (Pollack, 2005).

AI's application in cognitive assessment primarily focuses on two aspects: data collection and intelligent analysis, such as the use of machine learning models to interpret behavioral and physiological data to identify patterns, detect anomalies, and predict cognitive decline. Wearable devices, smart home systems, and mobile applications can continuously monitor the physiological and behavioral data of older adults, including gait, sleep, physical activity, and social behaviors. These data

accurately reflect daily activities and cognitive changes, enabling healthcare professionals to perform more precise cognitive assessments. Research has developed cognitive health assessment systems that combine wearable physiological sensors with environmental sensors, enabling automated cognitive health assessments within the natural living environments of older adults (Arif Ul Alam et al., 2020). AI systems can also analyze changes in gait, walking speed, and response time to help detect potential cognitive decline or other issues (Obuchi et al., 2024).

2.3.2 Cognitive Intervention

Cognitive intervention is another important application of AI in cognitive support for older adults. AI not only assesses cognitive health, but it also tailors cognitive training and intervention plans based on the assessment results. These plans are designed to help older adults maintain or enhance cognitive function and slow cognitive decline through personalized training.

AI-driven cognitive intervention systems use machine learning algorithms to automatically adjust the content and difficulty of training based on an older adult's cognitive level, health status, and personal needs (Arango et al., 2025; Kitakoshi et al., 2015). These cognitive exercises are often gamified to increase engagement and interest among older adults (Eun et al., 2022). For example, AI systems can design memory training, attention exercises, and problem-solving tasks to help older adults exercise their brains and enhance cognitive abilities (Kitakoshi et al., 2015). Through personalized, dynamically adjusted cognitive training, older adults can maintain challenge while avoiding frustration from tasks that are too difficult.

Furthermore, AI technologies can provide more flexible intervention forms through emotional analysis and natural language processing. Social robots and virtual assistants engage in daily conversations with older adults, providing cognitive training tasks alongside emotional support (Tokunaga et al., 2021). These interactive smart devices not only offer emotional companionship but also perform simple daily assistant tasks, such as medication reminders, playing music, and video calls (Magenat-Thalmann & Zhang, 2014). Studies have shown that social robot interventions significantly reduce older adults' depression, irritability, and anxiety scores (Yen et al., 2024). This multidimensional approach to cognitive intervention not only improves cognitive functioning but also provides greater emotional support.

2.3.3 Cognitive Recovery

Cognitive recovery is an advanced application of AI in cognitive support for older adults, focusing on restoring or improving cognitive abilities that have declined due to aging or disease (Vogan et al., 2020). Unlike traditional cognitive interventions, cognitive recovery aims to use concentrated and systematic training methods to restore as much cognitive function as possible and help older adults regain their ability to live independently (Hill et al., 2000).

AI applications in cognitive recovery primarily involve integrating virtual reality (VR) and augmented reality (AR) technologies (Catania et al., 2024). Virtual reality provides older adults with an immersive cognitive training experience, allowing them to exercise memory, spatial awareness, and problem-solving abilities through interactive games, simulated tasks, and virtual environments (García-Betances et al., 2015). AI-based cognitive training devices for older adults can flexibly adjust the difficulty of training based on the individual's cognitive condition, ensuring the challenge is appropriate while preventing undue psychological burden (Cai et al., 2023). These virtual training systems significantly enhance older adults' engagement and motivation while improving cognitive functioning (Bapka et al., 2018).

Additionally, AI-driven social robots are also essential tools in cognitive recovery. These robots, through anthropomorphic interaction patterns, assist older adults in effective cognitive recovery training (Pino et al., 2020). Unlike traditional rehabilitation training, social robots enhance older adults' social skills and cognitive flexibility through language and emotional communication (Góngora Alonso et al., 2019). This training not only helps improve cognitive abilities but also reduces feelings of loneliness and anxiety, promoting mental health (Lim, 2023; Hutson et al., 2011).

By combining personalized training plans, virtual reality, and social interaction, AI applications in cognitive recovery not only improve cognitive functioning but also promote greater engagement and accomplishment among older adults, supporting more effective recovery outcomes (Li & Liao, 2025). As technology continues to advance, AI's potential for cognitive recovery will grow, providing more comprehensive and effective cognitive support for older adults.

In summary, AI's applications in cognitive support for older adults primarily focus on assessment, intervention, and recovery. With precise cognitive assessment technologies, AI can facilitate early diagnoses and health management services; through personalized cognitive interventions, AI helps maintain and enhance cognitive function, effectively slowing cognitive decline; and through cognitive recovery systems, AI enables systematic training to maximize the restoration of cognitive abilities. As AI technology continues to progress, its applications in cognitive health management will become more widespread, helping older adults improve their quality of life and achieve healthier aging.

3 Potential Risks of AI in Elderly Care

3.1 Cultural Bias and Diversity Considerations in AI for Elderly Care

Cultural differences are significant factors influencing the quality of care when applying AI in elderly care (Li & Xu, 2021). For example, AI applications in elderly health management, daily care, and emotional support must consider the needs and

preferences of older adults from different cultural backgrounds to ensure more personalized and effective care (Huang et al., 2019). Older adults from other cultures have varying care needs, health beliefs, and lifestyles, which necessitate that AI systems be designed with cultural adaptability in mind to ensure their effectiveness and acceptance (Sgorbissa et al., 2018). In addition, health outcomes themselves may be shaped by culture, race, and ethnicity—leading to disparities that must be acknowledged in AI system design. Without intentionally including diverse data, AI algorithms risk producing biased recommendations that fail to meet the needs of underrepresented populations.

However, these cultural factors are not adequately addressed. In that case, AI systems may inadvertently embed cultural bias—such as prioritizing Western medical paradigms, communication norms, or care preferences—at the expense of more diverse or non-dominant cultural practices. These biases can manifest in mismatched care recommendations, reduced trust and usability among minority populations, or even exclusion from AI-enabled services. As such, addressing cultural bias is not only a design issue but a matter of equity and justice in aging services.

First, cultural differences significantly impact the care needs of older adults. In some cultures, older adults are more likely to rely on family members for care, with family members often assuming responsibility for health management rather than seeking professional external care (Liu, 2000). This reliance on family-based caregiving can sometimes conflict with the personalized care recommendations provided by AI systems (Chen & Hou, 2023). To address this, AI technology needs to be designed to align with family care models. Furthermore, some cultures prefer traditional medicine to modern medical practices, so AI care systems must respect these traditional health beliefs and integrate both conventional and contemporary medical knowledge. Equally important, in many cultures, human contact is itself a valued part of care. Emotional connection, physical presence, and interpersonal trust are deeply embedded in caregiving traditions, and AI should be positioned as a supplement rather than a replacement for the human element.

Moreover, the acceptance of AI technology varies across cultures. In some cultures, older adults generally exhibit low acceptance of modern technology, especially when they are not proficient in its use. AI technology is often viewed as difficult to trust and cumbersome to operate (Cotten et al., 2024). Therefore, AI systems need to focus on simplifying user interfaces, reducing complexity, and providing multi-language support to ensure that all older adults can use these technologies more easily.

In conclusion, applying AI in elderly care requires not only sensitivity to cultural diversity but also critical awareness of potential cultural bias embedded in technology. Social workers and technologists must work collaboratively to ensure that AI systems are culturally inclusive, respectful of diverse values, and capable of delivering equitable care to all older adults—regardless of cultural background.

3.2 *Privacy Concerns in AI for Elderly Care*

As AI becomes increasingly integrated into elderly care, privacy protection and data security have emerged as critical ethical challenges. While real-time monitoring and personalized care improve efficiency and responsiveness, they also involve extensive data collection that may compromise the privacy of older adults (Bartoletti, 2019; Htet et al., 2024). Many older adults appreciate the benefits of AI-assisted technologies but remain concerned about potential misuse of their personal information (Hu & Chen, 2024).

To ensure trust and ethical integration, AI systems must balance effective data use with transparency, security, and respect for personal control. This section explores three major dimensions of privacy in AI for elderly care—each carrying significant implications for social work: informed consent and transparency, data security and risk prevention, and autonomy and ethical governance.

3.2.1 Informed Consent and Data Transparency

The application of AI in elderly care requires the collection and use of large amounts of sensitive personal data, including physiological indicators, behavioral patterns, and emotional states. Suppose data collection and use processes are not transparent. In that case, older adults and their families may lack clarity about how their data is handled, for how long it is stored, and whether it might be shared or used for commercial purposes (Huang et al., 2019). This lack of transparency can erode trust and disempower users.

To respect privacy rights, AI developers and care providers must establish clear, accessible, and culturally appropriate informed consent processes (Friedman et al., 2022). These processes should explain in plain language who will have access to health data, how it will be used, and under what circumstances it might be shared or revoked. This is a critical area where social workers play a key role in advocating for and facilitating meaningful informed consent for older adults and their caregivers. In addition to obtaining informed consent, ongoing education and support are essential to help older adults and their caregivers understand how AI technologies function, the potential risks involved, and how they can exercise control over their personal data. This educational component is critical to empowering users and ensuring that consent is documented and meaningfully informed.

3.2.2 Data Security and Risk Prevention

AI systems often rely on cloud platforms and large-scale data processing, which introduces significant cybersecurity risks (Murdoch, 2021). Health data is particularly sensitive and valuable, making it a prime target for breaches and cyberattacks. For example, a significant data leak in 2024 exposed millions of health records,

including information on elderly individuals and their caregivers (Wan, 2024; Alder, 2025).

Older adults may be especially vulnerable due to limited digital literacy or awareness of cybersecurity risks. Therefore, it is essential to implement strong encryption protocols, multi-layered authentication, and regular security audits to safeguard data integrity. These measures should be paired with user education and institutional accountability to reduce vulnerabilities and prevent misuse.

3.2.3 Autonomy and Ethical Governance

Beyond transparency and technical safeguards, AI systems must uphold the ethical principle of autonomy. As AI becomes more involved in recommending or executing care decisions, there is a risk that older adults—particularly those with cognitive decline—may lose meaningful control over their personal data and decisions (Mukaetova-Ladinska et al., 2020; O'Connor, 2022).

Ethical governance should ensure that older adults retain decision-making power over how their data is used and shared. When individuals are no longer able to exercise this control, mechanisms such as digital advance directives or designated data proxies can help ensure their wishes are respected (Duque & Otaegui, 2023). Social workers can act as key intermediaries in this process—interpreting AI-generated insights, ensuring alignment with the individual's values, and safeguarding ethical boundaries.

AI-related privacy concerns in elderly care are multifaceted—spanning transparency, technical security, and ethical control. Addressing them requires not only technological safeguards but also inclusive and culturally competent legal-ethical frameworks. Social workers are essential actors in this process, helping older adults navigate complex systems, advocating for informed consent, and ensuring that privacy and autonomy remain central to AI-enabled care.

3.3 Technology Dependence in Elderly Care

As AI becomes more widely used in elderly care, the issue of technology dependence is increasingly becoming a significant concern. While AI has improved the efficiency and convenience of care, excessive reliance on technology has raised concerns about the loss of autonomy for older adults, increased risk of technical failures, and potential social isolation (Salomé & Monfort, 2023). Moreover, as older adults become more reliant on AI systems, the risk of digital exploitation, fraud, or abuse increases—particularly when safeguards and oversight are lacking.

3.3.1 Impact of Technology Dependence on Autonomy

In elderly care, although AI provides conveniences such as health monitoring and medication management, over-reliance on technology may undermine older adults' autonomy (Duque & Otaegui, 2023). While smart devices and virtual assistants offer numerous benefits, they can inadvertently weaken older adults' ability to manage their own health and make independent decisions actively (Dahlke & Ory, 2017). For instance, if older adults overly rely on AI systems for medication reminders, they may become less engaged with their health, leading to a decrease in their sense of autonomy and, ultimately, a reduction in quality of life (Stypińska & Franke, 2023).

3.3.2 Risks of Technical Failures and Dependence

As reliance on AI technology increases, the risks associated with technical failures and operational errors also rise (Arnold & Toner, 2021). Older adults often lack proficiency with technology, and if AI health devices malfunction or data is lost, health warnings may be overlooked, delaying critical treatments (Huang et al., 2019). For example, if monitoring devices fail, older adults may not receive timely health alerts or instructions, increasing the risk of delayed medical response and adverse health outcomes (Illario et al., 2016). The occurrence of technical failures often leaves older adults feeling helpless, deepening their dependence on technology (Claßen, 2013).

3.3.3 Potential for Social Isolation Due to Technology Dependence

Although AI can help older adults maintain certain social connections through virtual companionship and remote healthcare, prolonged reliance on technology may reduce face-to-face interactions with family and friends, leading to social isolation (Khosravi et al., 2016). Virtual interactions lack emotional depth and warmth and cannot replace genuine human social experiences. Prolonged reliance on AI for communication and health management may lead older adults to become more isolated, negatively affecting their mental and emotional health and their sense of social participation (Latikka et al., 2021).

In conclusion, while AI technology in elderly care offers significant conveniences, over-dependence can lead to diminished autonomy, increased risks from technical failures, and heightened social isolation. Therefore, AI care solutions should balance technological support with interpersonal interaction, focus on fostering older adults' independence, reduce the risk of technical failures, and ensure that older adults maintain social and emotional connections.

4 Designing Ethical AI Care Systems that Respect the Autonomy and Dignity of Older Adults

As the global population ages, demand for elderly care continues to rise, while traditional care models struggle to meet increasingly complex needs. In this context, AI has demonstrated significant potential—offering innovative solutions that support daily functioning, health monitoring, and personalized care for older adults.

However, integrating AI into elderly care must go beyond technical efficiency. To truly serve aging populations, AI systems should be guided by humanistic principles—especially the values of autonomy and dignity, which are foundational to social work practice. These principles, embedded in the Social Work Code of Ethics, emphasize respect for older adults' rights, preferences, and lived experiences.

Designing ethical AI care systems that uphold these values not only enhances quality of life but also fosters independence and promotes healthy aging (Hwang et al., 2025). As such, this section explores how ethical, social, and technological considerations can be integrated to ensure that AI innovations align with the human-centered mission of social work.

4.1 Human-Centered Design and Ethical Considerations

When applying AI technology in elderly care, it is crucial to comprehensively address ethical issues, particularly those related to privacy, consent, and bias elimination (Iqbal, 2023). AI should be viewed as a semi-autonomous participant capable of influencing older adults' functional and social relationships. Therefore, when designing AI care technologies, it is essential to focus on human-centered attributes, values, and design principles to ensure they meet ethical standards and align with societal needs (Dignum, 2017). The goal of ethical design is to ensure that technology supports rather than undermines the freedom and dignity of older adults, helping them maintain autonomy while respecting their values (Zhou & Salvendy, 2019).

4.2 AI Supporting Independent Living for Older Adults

AI technology provides strong support for older adults in maintaining independent living, especially in areas such as health monitoring, accident risk alerting, and personalized care. With real-time monitoring and precise health status detection capabilities, AI can seamlessly integrate into older adults' daily lives and provide appropriate support. For example, AI can play an essential role in elderly health management by monitoring physiological data such as heart rate, blood pressure, and oxygen saturation levels and using wearable devices or sensors to provide early warnings for abnormal health conditions (Yan et al., 2015). Additionally, AI can

offer personalized care recommendations, such as predicting potential health risks based on physiological parameters and assisting in developing personalized rehabilitation plans (Puaschunder & Feierabend, 2019). For example, AI has been used in fall detection and Alzheimer's disease prevention, achieving more than 98% accuracy through technologies such as thermal imaging sensors and millimeter-wave radar, enabling timely health interventions (Li, 2024). This AI design approach not only enhances the usability of technology but also increases older adults' sense of participation, reducing their dependence on others, boosting their confidence, and improving their quality of life (Luo et al., 2025).

4.3 Maintaining Autonomy and Dignity

Protecting the autonomy and dignity of older adults is a core element in AI care design (Mhlanga, 2023). In elderly care, healthcare professionals should work with older patients and their significant others to ensure that the wishes and needs of older adults are fully considered in care decision-making processes (Aita, 2013). This not only fosters the development of personalized care decisions but also enhances the autonomy and sense of dignity of older adults, thereby improving their health and quality of life (Nikolidakis et al., 2010). AI systems should allow older adults to participate in care decisions and avoid excessive interference in their daily lives (Zhu et al., 2022). Ensuring that older adults feel fully respected and supported in the caregiving process has a positive impact on their overall well-being (Czaja & Ceruso, 2022). In the design process, particular attention should be given to avoiding over-dominating technology to ensure that older adults still retain control over their health care decisions. By integrating user-centered design, AI technologies can better meet the needs of older adults for independent living while maintaining their dignity and autonomy (Wang et al., 2019).

4.4 Multidisciplinary Collaboration and Participatory Design

The successful application of AI technology in serving older adults also depends on multidisciplinary collaboration, including the involvement of clinical doctors, psychologists, ethicists, and engineers (Wu et al., 2018). Cross-disciplinary partnership helps ensure that the practical application of technology aligns with ethical and professional standards (Yigit et al., 2025). Additionally, older adults' participation in the design, implementation, and evaluation of AI systems is crucial. Incorporating the needs and real-life feedback from older adults is essential to addressing challenges such as the digital divide and inequality, ensuring that AI system designs truly align with their daily experiences and evolving care needs (Demiris, 2024). This participatory approach reflects the principles of participatory action research,

which centers the voices of those most affected and promotes more equitable, responsive system design.

As shown, designing AI care technologies that respect the autonomy and dignity of older adults requires a comprehensive, inclusive approach that considers ethics, human-centered and user-centered design, and multidisciplinary collaboration. Through these methods, AI technology can effectively support older adults in maintaining independent living and enhance their quality of life and well-being. As technology continues to develop, AI care systems will play an increasingly important role in promoting healthy aging while respecting the autonomy and dignity of older adults.

5 Conclusion

As the global aging population continues to grow, the demand for elderly care is increasing, and traditional caregiving models face numerous challenges. The application of AI in elderly care offers innovative solutions, particularly in health monitoring, social support, and the maintenance of cognitive function, demonstrating immense potential. AI not only significantly improves the efficiency of caregiving but also enhances personalized services, enabling older adults to make more informed choices. This, in turn, greatly strengthens their ability to live independently and significantly improves their overall quality of life.

However, there remains a gap between the widespread adoption of AI technology and its adaptability among older adults. This may affect their ability to use AI technologies effectively. Furthermore, during the design phase, AI technology needs to balance ease of use and intuitive functionality to reduce technological barriers and improve quality of life. While AI can offer some emotional support, it is inherently limited in replicating the depth and authenticity of human interaction—and from a humanistic perspective, it should not be expected to do so thoroughly.

Additionally, as AI technology becomes more widely applied in the caregiving field, privacy protection and ethical concerns are becoming increasingly important. Data security and algorithmic ethics have become crucial factors in balancing technological innovation with the protection of personal privacy. When AI systems analyze patient data to support diagnosis, it is essential to ensure that data collection and processing adhere to strict privacy principles. Furthermore, the development and application of AI algorithms must respect individual privacy, proactively address algorithmic bias, and ensure interpretability and transparency. If left unaddressed, algorithmic bias can compromise both the accuracy of AI decision-making and the effectiveness of care recommendations, particularly for underserved or underrepresented older adults. Ensuring fairness in AI outcomes is therefore a fundamental part of building trustworthy and inclusive AI systems for aging populations.

To address these issues, future research should further focus on the adaptability of AI systems, cultural differences, ethical design, and privacy protection. Moreover,

AI should be integrated with human caregiving to ensure that technology does not excessively interfere with older adults' autonomy and social interaction. Through multidisciplinary collaboration and technological optimization, AI will become more widespread in elderly care, offering more comprehensive and personalized caregiving services to older adults and ultimately achieving the goal of healthy aging.

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